

Element A: **Presentation and Justification of the Problem**

The Amur leopard is a leopard subspecies native to the Primorye region of southeastern Russia and northern China, where they get cognitive stimulation through the complex, unpredictable, and physically demanding nature of their environment. In the wild, Amur leopards roam large territories, hunt prey, track scents, and constantly navigate varied terrain. These activities provide consistent mental and physical stimulation that supports their behavioral and psychological health (World Wildlife Fund).

A female Amur Leopard named Natasha (12 years old, ~35–39 kg) lives at the Nashville Zoo, along with another female named Roza (4 years old, ~35 kg) and a male named Jibari (5 years old, ~55 kg). While in captivity, Natasha, along with the others, experiences a far more controlled and predictable environment. Although zoos attempt to replicate natural habitats, captive environments still contain fewer novel stimuli than the wild. When animals lack stimulation, they may develop stereotypic behaviors, sometimes referred to as zoochosis. These behaviors may include pacing, inactivity, or repetitive movement patterns caused by boredom or stress (Vashisth et al.). Research shows that captive leopards without adequate enrichment display significantly more inactivity and stereotypic behaviors compared to those provided with enrichment activities (Vashisth et al.).

Therefore, the problem is that current enrichment methods at the Nashville Zoo do not consistently engage Natasha the Amur leopard for extended periods of time, limiting her cognitive stimulation and preventing the full expression of natural hunting and foraging behaviors.

However, studies have proven that olfactory or sensory stimuli have positive effects on Amur leopards and other big cats. For example, one study observed that providing scent-based enrichment increased behavioral diversity and exploratory activity in captive Amur leopards by introducing new smells such as prey odors or spices (Yu et al.). Similarly, other research found that environmental enrichment—particularly sensory stimuli like scents—can increase activity levels and encourage investigative behaviors in captive felids (McWhorter). These findings suggest that enrichment tools that incorporate scent, puzzle solving, or hidden food rewards can significantly improve animal welfare in zoo environments.

The Nashville Zoo opened a new Amur leopard exhibit in 2025 and continues to work toward providing enrichment that mimics the cognitive challenges the animals would face in the wild. Nashville zookeepers are constantly trying to find ways to entertain the leopards with various enrichment activities and toys, offering activities each day. However, enrichment objects can quickly become repetitive. Once the novelty of a toy or activity wears off, the leopards often lose interest and stop interacting with it. Research indicates that enrichment stimuli may lose effectiveness rapidly, with interaction decreasing significantly after the first exposure if the stimulus remains unchanged (McWhorter).

According to Natasha's primary zookeeper, Natasha particularly enjoys puzzle-based enrichment that requires her to work through challenges to obtain food rewards. Zookeepers report that she is highly intelligent and motivated by tasks that require problem solving and persistence. Because of this, enrichment devices that incorporate multiple steps or puzzles may increase the amount of time she engages with enrichment activities. Currently, enrichment interactions may only last short periods of time. Zookeepers aim for activities that engage Natasha for at least 5–20 minutes, with an ideal engagement time of around 30 minutes.

Increasing the time she spends actively working for enrichment would help stimulate natural behaviors such as hunting, foraging, and exploring.

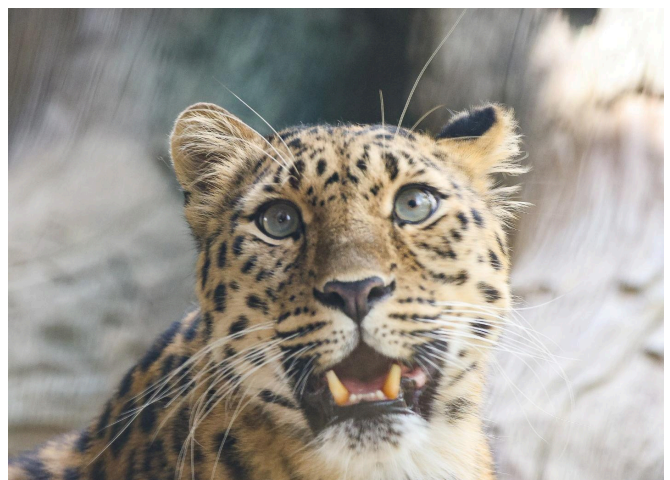
Food rewards also play an important role in enrichment. Like many animals, Natasha is motivated by food, including whole prey items such as rabbits, birds, and fish. Whole prey feeding is particularly stimulating because it encourages natural hunting and foraging behaviors and often results in the longest engagement times. Replicating similar problem-solving behaviors through enrichment devices could help extend these natural behaviors within a captive environment.

Several stakeholders are directly connected to this problem. The primary stakeholder is Natasha herself, whose mental and physical well-being depends on daily cognitive stimulation and enrichment. Zookeeper A, Natasha's primary zookeeper, is also a major stakeholder because she is responsible for Natasha's daily care and enrichment planning. Secondary stakeholders include Roza and Jibari, since they can also use the enrichment toy and will benefit greatly from it. Zoo veterinarians are also secondary stakeholders because the physical and psychological health of the leopards falls under their responsibility. Zoo managers and administrators are also secondary stakeholders because they must ensure that the zoo meets modern animal welfare standards while maintaining engaging exhibits for visitors. Additionally, wildlife conservation organizations such as Wildlife Vets International are tertiary stakeholders because captive breeding and conservation programs rely on maintaining healthy and behaviorally normal animals that may contribute to species survival efforts. Another tertiary stakeholder is the zoo visitors, who wish to continue seeing the leopards and will benefit from the betterment of their wellbeing.

The concerns of these stakeholders emphasize the importance of addressing the enrichment challenge. Without sufficient stimulation, captive animals may experience chronic stress, muscle atrophy from inactivity, or behavioral abnormalities that reduce their quality of life. For zookeepers and veterinarians, ineffective enrichment tools create ongoing challenges in maintaining animal welfare. Zoo administrators must also consider visitor education and conservation messaging, as zoos play a significant role in raising awareness about endangered species like the Amur leopard. According to conservation organizations, the Amur leopard is one of the rarest big cats in the world, with fewer than 100 individuals remaining in the wild (World Wildlife Fund).

Addressing enrichment challenges is therefore critical not only for Natasha's daily welfare but also for broader conservation goals. Developing a novel enrichment device that engages Natasha for longer periods of time would improve her cognitive stimulation, reduce stereotypic behaviors, and help zoo staff provide more effective enrichment strategies. Since enrichment devices often lose effectiveness quickly, designing a puzzle-based toy that incorporates scent and food rewards could provide longer engagement and improve animal welfare. Solving this problem would benefit Natasha directly while also supporting the zoo's goals of conservation, education, and ethical animal care.

Photos of Natasha:





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